7 Work out

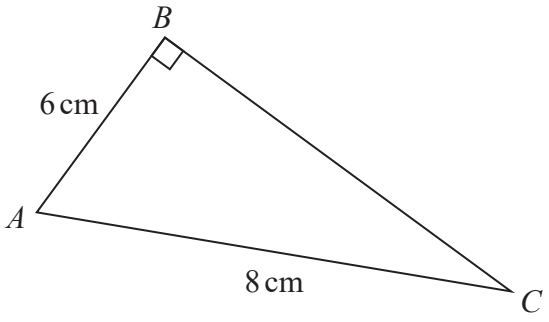
$$\sqrt{\frac{2.5 \times \sin 43°}{8.2^2 - 50.5}}$$

Give your answer correct to 3 significant figures.

.....

(Total for Question 7 is 2 marks)

8 ABC is a right-angled triangle.



Here is Sarah's method to find the length of BC.

$$\begin{aligned} BC^2 &= AB^2 + AC^2 \\ &= 6^2 + 8^2 \\ &= 100 \\ BC &= 10 \end{aligned}$$

(a) What mistake has Sarah made in her method?

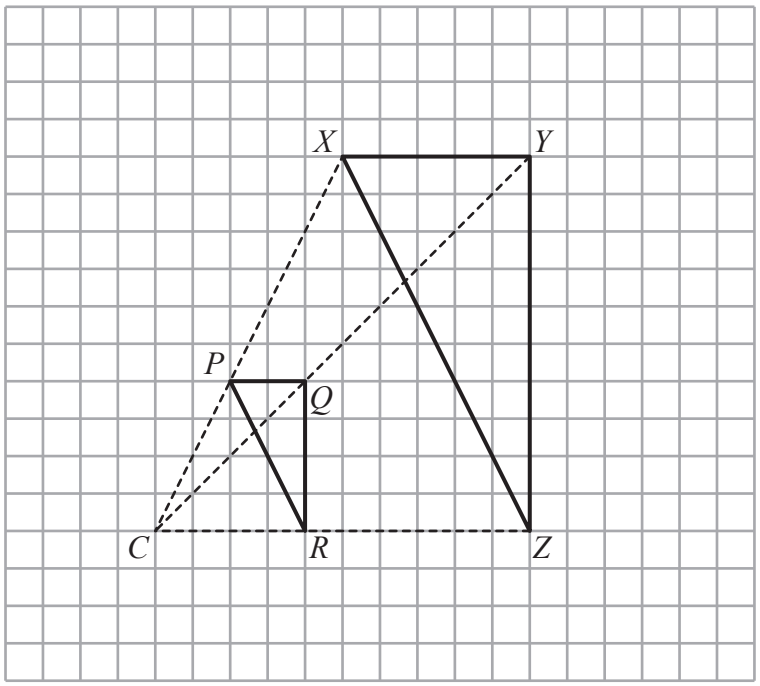
AC = 8 is the hypotenuse.

$BC^2 = 8^2 - 6^2$, not $8^2 + 6^2$.

1 mark
Subtract the shorter-side square.

(1)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA



Roy is going to enlarge triangle PQR with centre C and scale factor $1\frac{1}{2}$

He draws triangle XYZ.

(b) Explain why Roy's diagram is **not** correct.

Correct 1½: 2 original squares → 3 new.

Roy drew 5 new squares: factor 2½.

~~1 mark~~
Roy used 2½, not 1½.

(1)

(Total for Question 8 is 2 marks)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA